Targeted SAT Math — Set 04

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Homework after Bluebook Practice Test 4. Work on paper. Calculator / Desmos is fine.

These are **new** questions on the four Module 2 items that missed — not those official questions, and not copies from Set 01.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

Write the job first.

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1. Notes for this set only

Exponential period — one cycle = exponent equals 1

A model $A(1+r)^{kt}$ finishes **one** growth (or decay) cycle when the exponent equals **1**.

- The **percent** n comes from the **base**. 1.08 means $n = 8$. It is not 1.08, and it is not a leftover number in the fraction.
- The coefficient k and the period **in years** are **reciprocals**. If one cycle takes $\frac{3}{2}$ years (18 months), then $k = \frac{2}{3}$.
- If they hide the exponent and give you the percent and the months: convert months → years, then $k = 1/(\text{years per cycle})$.
- If t is already in **months**, do not also divide by 12.
- Sometimes the exponent is written unsimplified, like $t/0.75$. Same clock. Set that expression equal to 1.

Add a constant to every data value

- **Center moves:** mean, median, and quartiles each change by that constant.
- **Spread stays:** range, IQR, and standard deviation do **not** change. The list is only translated.
- Read which two stats they named. Median and range is not the same question as mean and range.

Rational $f(x) = \dfrac{a}{x+b}$, then a shift

- Vertical asymptote $x = -b$. One point on the curve gives a.
- $g(x) = f(x+h)$ means **replace** x by $x+h$. Do not stop at the equation for f.
- $g(x) = f(x-h)$ is the other direction. Still replace, then simplify.

Parabola — vertex, product, sum

- $a+b+c$ is the y-value at $x = 1$. Write vertex form and plug in $x = 1$.
- Two x-intercepts and a vertex **below** the axis $\Rightarrow$ opens up $\Rightarrow$ $a > 0$, so $f(1)$ is **greater** than the vertex y.
- For $ax^2 + bx + c = 0$: **sum** of solutions is $-\dfrac{b}{a}$. Keep the minus. **Product** is $\dfrac{c}{a}$.
- If they ask for the product of the sum and the product, that is $\left(-\frac{b}{a}\right) \times \frac{c}{a}$. Do both, then multiply.
- If they write the product as k times an expression, k is whatever is left after c/a.

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2. Questions

1. The function P is defined by $P(t) = 420(1.12)^{\frac{3}{5}t}$, where t is the number of years after 2016. According to the model, the population increases by $n\%$ every 20 months. What is the value of n?

A) 3
B) 5
C) 12
D) 20

2. Data set A is $8, 10, 10, 11, 15$. Data set B is created by adding 7 to each value in data set A. Which statement is true?

A) The median of B equals the median of A, and the range of B equals the range of A.
B) The median of B equals the median of A, and the range of B is greater than the range of A.
C) The median of B is greater than the median of A, and the range of B equals the range of A.
D) The median of B is greater than the median of A, and the range of B is greater than the range of A.

3. In the xy-plane, a parabola has vertex $(4, -10)$ and intersects the x-axis at two points. If the equation is written as $y = ax^2 + bx + c$, which of the following could be the value of $a + b + c$?

A) -19
B) -13
C) -10
D) -7

4. A bacteria culture increases by 8% every 18 months. The function C models the size of the culture t years after 2019. Which equation could define C?

A) $C(t) = 80(1.08)^{\frac{2}{3}t}$
B) $C(t) = 80(1.08)^{\frac{3}{2}t}$
C) $C(t) = 80(1.08)^{18t}$
D) $C(t) = 80(1.08)^{t/18}$

5. The rational function f is defined by $f(x) = \dfrac{a}{x+b}$, where a and b are constants. The graph of $y = f(x)$ has a vertical asymptote at $x = -2$ and passes through $(0, 3)$. If $g(x) = f(x - 3)$, which equation could define g?

A) $g(x) = \dfrac{6}{x+2}$
B) $g(x) = \dfrac{6}{x-1}$
C) $g(x) = \dfrac{6}{x-5}$
D) $g(x) = \dfrac{6}{x}$

6. (SPR) A lake's algae count increases by 7% every 15 months. The count t years after 2021 is modeled by $A(t) = B(1.07)^{kt}$, where B and k are constants and $k > 0$. What is the value of k?

7. In the given equation, p and q are positive constants.

$$8x^2 + (8p + q)x + pq = 0$$

The product of the solutions is kpq, where k is a constant. What is the value of k?

A) $\frac{1}{8}$
B) $\frac{1}{4}$
C) 1
D) 8

8. The rational function h is defined by $h(x) = \frac{a}{x+b}$. The graph of $y = h(x)$ has a vertical asymptote at $x = 5$ and passes through $(7, 4)$. If $p(x) = h(x+2)$, which equation could define p?

A) $p(x) = \frac{8}{x-5}$
B) $p(x) = \frac{8}{x-3}$
C) $p(x) = \frac{8}{x-7}$
D) $p(x) = \frac{8}{x}$

9. (SPR) Let s be the sum of the solutions to $6x^2 + 18x + 5 = 0$ and let p be the product of the solutions. What is the value of sp?

10. The function S is defined by $S(t) = 110(1.10)^{t/0.75}$, where t is years after 2015. According to the model, the quantity increases by $n\%$ every 9 months. What is the value of n?

A) 0.75
B) 9
C) 10
D) 1.10